Detecting Premature Accountability Convergence: A Computational Framework for Regulatory Investigations

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Abstract Premature accountability convergence (PAC) occurs when investigative reasoning closes too early on a single explanation, despite available alternatives. While prior work has theorized this phenomenon in regulated environments, no computational framework exists for its detection. I present a feature-based classification approach that operationalizes PAC as a process-quality detection task, distinct from correctness assessment. Using a synthetic dataset (n=500) validated against an FDA Warning Letter corpus (N=50, 2016–2025), I train and evaluate three classifiers (Logistic Regression, Random Forest, SVM), achieving AUC scores above 0.97 across all models. Ablation analysis identifies commitment level as the strongest predictor (ΔF1 = 0.060), while SHAP analysis reveals that explanation count drives case-level variation. Our framework provides a foundation for accountability-aware AI systems in high-stakes regulatory contexts.

Keywords: accountability, premature convergence, NLP, regulatory AI, FDA, explainability

Introduction

In high-stakes investigative settings, accountability is not simply discovered from evidence — it is constructed through sequential interpretive decisions. When this process closes too early on a single explanation, premature accountability convergence (PAC) occurs: alternatives remain available, but reasoning has already stabilized (1).

PAC is particularly consequential in regulated manufacturing environments, where root cause analysis (RCA) determines corrective actions, regulatory responses, and organizational liability. A premature conclusion — even a correct one — may leave systemic contributors unexamined, increasing the risk of recurrence.

Despite growing interest in accountability in AI and organizational systems (3), no computational framework exists for detecting PAC as it occurs. Prior work has theorized convergence as a social and organizational phenomenon, but has not operationalized it as a detectable signal in text or process data.

This paper makes three contributions:

- I formalize PAC detection as a *process-quality* classification task, distinct from correctness assessment.
- I design an entropy-based explanatory diversity score and a feature set grounded in PAC theory.
- I validate our framework against an FDA Warning Letter corpus (N=50, 2016–2025), demonstrating empirical alignment between synthetic and real-world regulatory data.

Related Work

Accountability in Computational Systems

Accountability has been studied extensively in public administration and AI governance (3), but computational approaches remain limited. Prior work on algorithmic accountability has focused on outcome fairness (4) and transparency (5), rather than the *process* by which responsibility is constructed. I extend this line of work by treating accountability convergence as a computationally tractable phenomenon.

Convergence and Closure in Reasoning

Premature closure — the tendency to settle on an explanation before exhausting alternatives — has been documented in medical diagnosis (6), legal reasoning, and organizational investigation. However, existing work relies on qualitative analysis or expert judgment. Our framework operationalizes premature closure as a detectable signal using structural and linguistic features.

NLP for Regulatory Text

Recent work has applied NLP to regulatory documents including clinical notes (7) and legal corpora. FDA Warning Letters have been used for compliance analysis (2), but not for process-quality assessment. I introduce this corpus as a validation resource for convergence detection.

Framework

Definition

I define premature accountability convergence at time t as follows:

> *Premature convergence at time t occurs when the reasoning state exhibits low explanatory diversity and high attribution commitment, despite later evidence demonstrating that additional plausible explanations or attributional revisions were still available (1).*

This definition has three key properties. First, it is *process-oriented*: correctness of the final conclusion is irrelevant. Second, it is *temporally grounded*: convergence is assessed relative to what was knowable at time t. Third, it is *empirically operationalizable*: each component can be measured from process data or document structure.

Feature Design

I operationalize PAC through six features derived from reasoning logs or investigative documents:

- **Explanation count**: number of distinct causal explanations considered at time t
- **Diversity score**: Shannon entropy over explanation categories, $H = -\sum p_i \log_2 p_i$
- **Commitment level**: ordinal scale (0=low, 1=medium, 2=high) reflecting epistemic certainty
- **Uncertainty marker**: presence of hedging language (*possibly*, *may*, *unclear*)

- **Closure language**: presence of commitment language (*clearly*, *obviously*, *root cause is*)
- **Alternative considered**: whether alternative explanations were explicitly examined

Labeling

Labels are assigned probabilistically via a risk score:

$$r = w_1(3-e) + w_2 c + w_3(1-u) + w_4 l \tag{1}$$

where e, c, u, l denote explanation count, commitment level, uncertainty marker, and closure language respectively, and $w_1 = w_2 = w_3 = w_4 = 0.25$. Gaussian noise $\mathcal{N}(0, 0.15)$ is added to simulate real-world label ambiguity. Cases exceeding threshold $\theta = 0.5$ are labeled as premature ($y = 1$).

Experiments

Dataset

I construct a two-part dataset following a synthetic-to-real validation design.

Synthetic dataset. I generate $n = 500$ cases using probabilistic labeling (Section 3.3), yielding a near-balanced distribution (42% premature, 58% non-premature). The mean explanatory diversity score is 1.584 (SD = 0.821).

FDA Warning Letter corpus. I validate against $N = 50$ FDA Warning Letters (2016–2025, 21 CFR 210/211), comprising 160 violations across 50 finished pharmaceutical manufacturing cases (2). The corpus yields a mean diversity score of 1.601 (SD = 0.489), closely approximating the synthetic distribution ($\Delta = 0.017$), supporting the ecological validity of our synthetic design.

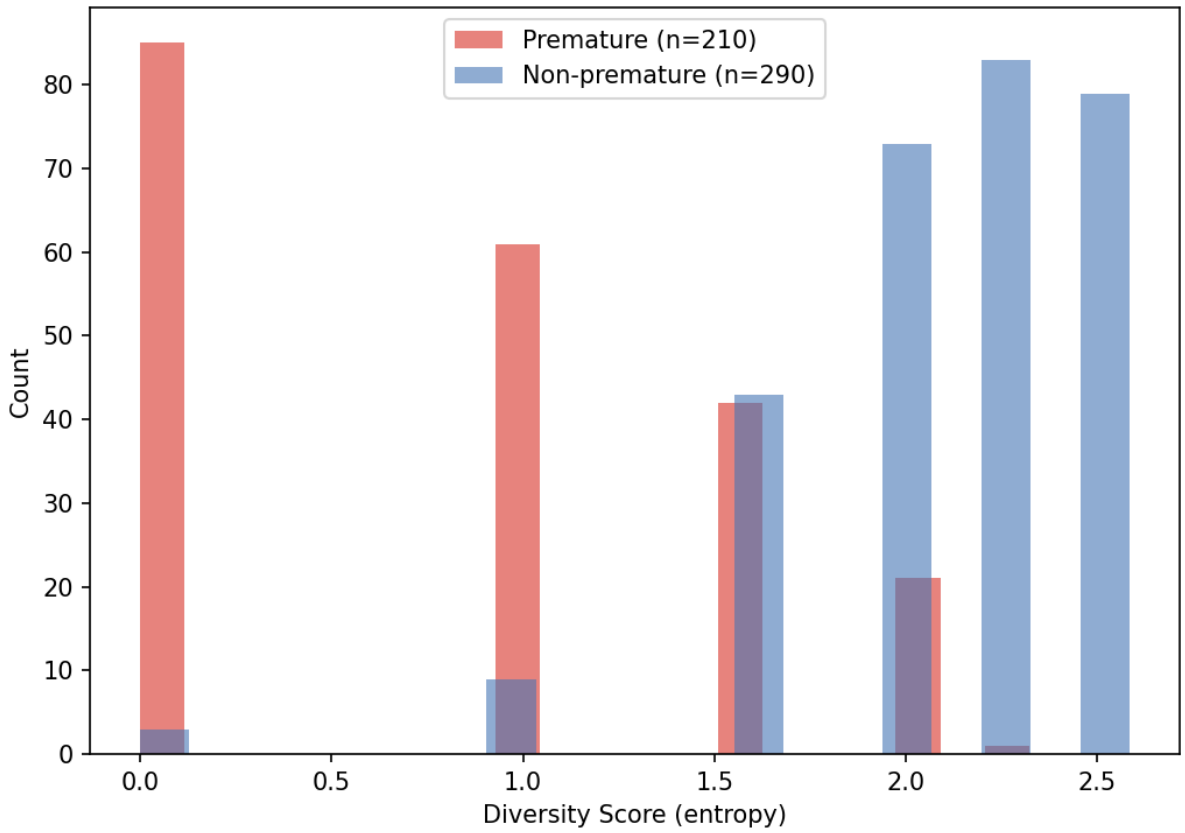


Figure 1: Explanatory diversity score distribution. Premature cases concentrate near zero entropy, while non-premature cases spread toward higher diversity.

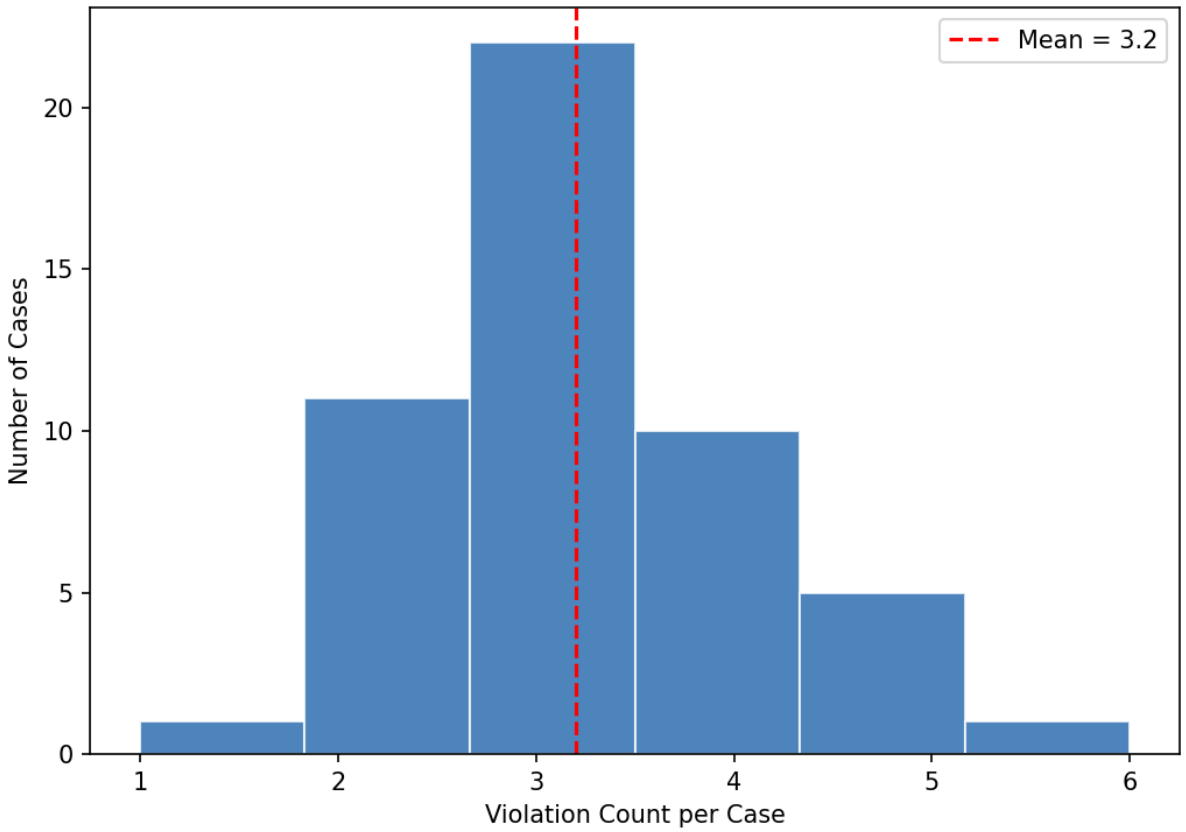


Figure 2: Violation count distribution across 50 FDA Warning Letter cases. Mean = 3.2 violations per case.

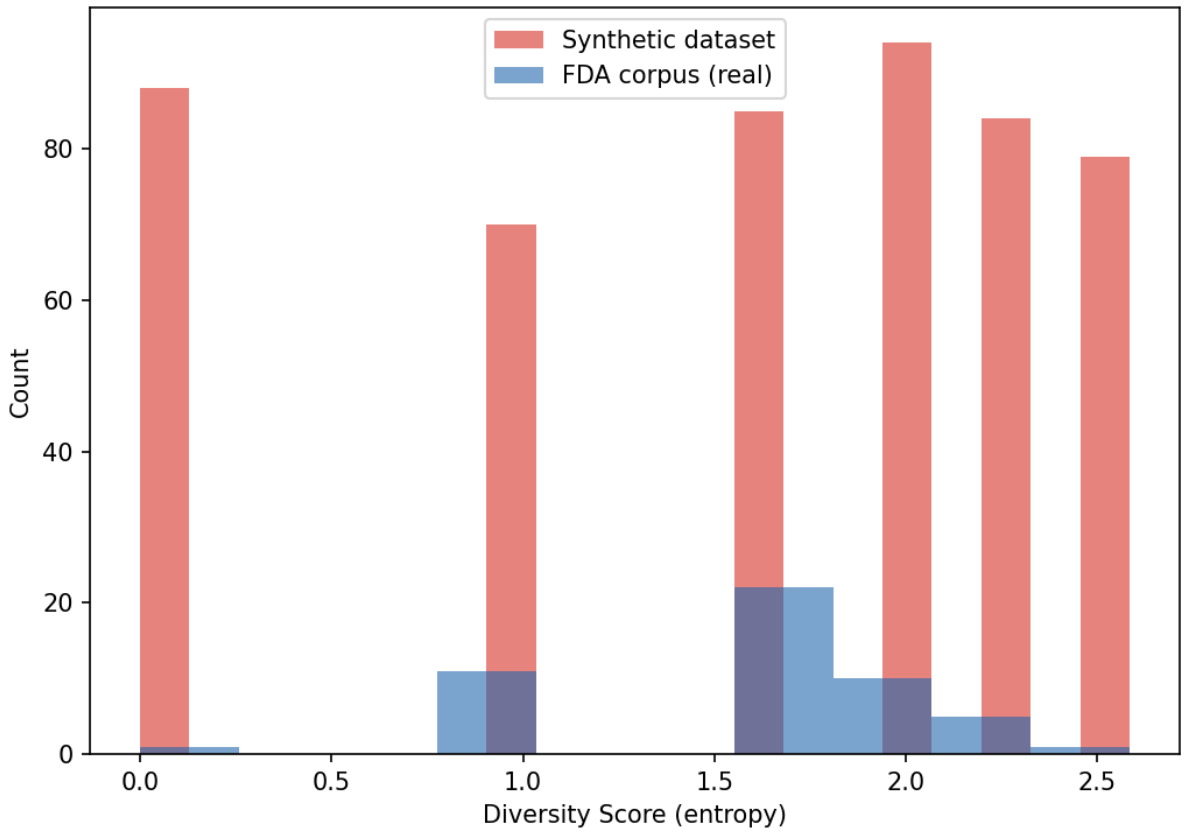


Figure 3: Diversity score distribution: synthetic dataset vs FDA Warning Letter corpus. Mean scores are nearly identical (1.584 vs 1.601).

Models

I evaluate three classifiers within a standardized pipeline (StandardScaler + classifier): Logistic Regression (LR), Random Forest (RF, $n = 100$ trees), and Support Vector Machine (SVM, RBF kernel). All models are evaluated via 5-fold cross-validation with F1 as the primary metric, and AUC on a held-out test set (20%).

Results

Table 1 reports model performance. All three classifiers achieve consistent F1 scores (0.901–0.913) and AUC above 0.97, suggesting that premature convergence is a robustly detectable signal rather than an artifact of any particular model architecture.

Table 1: Model Comparison (5-fold CV)

Model	**F1 (CV)**	**AUC**
Logistic Regression	0.909 ± 0.031	0.973
Random Forest	0.901 ± 0.031	0.975
SVM	0.913 ± 0.036	0.974

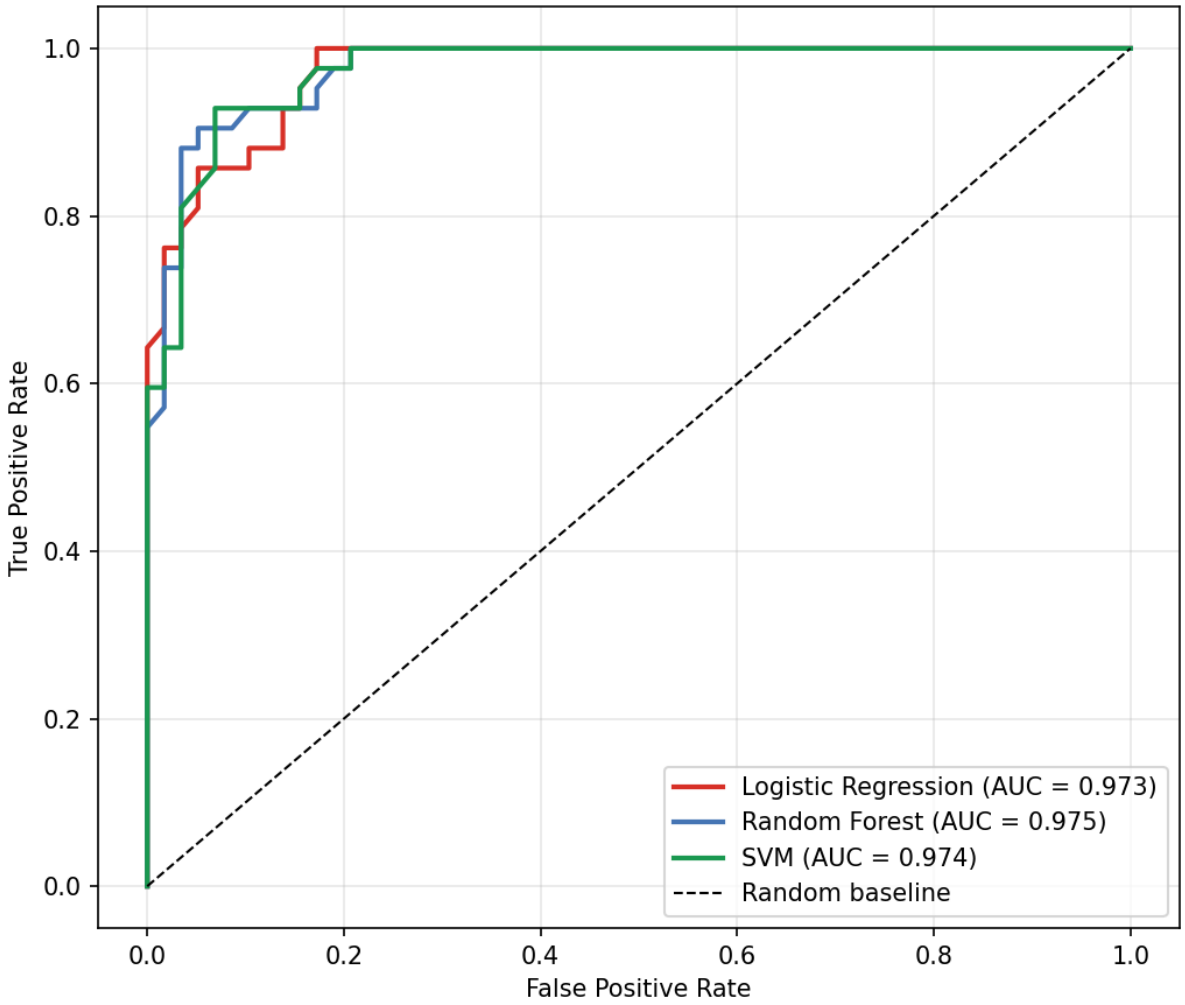


Figure 4: ROC curves for all three classifiers. All models achieve AUC above 0.97.

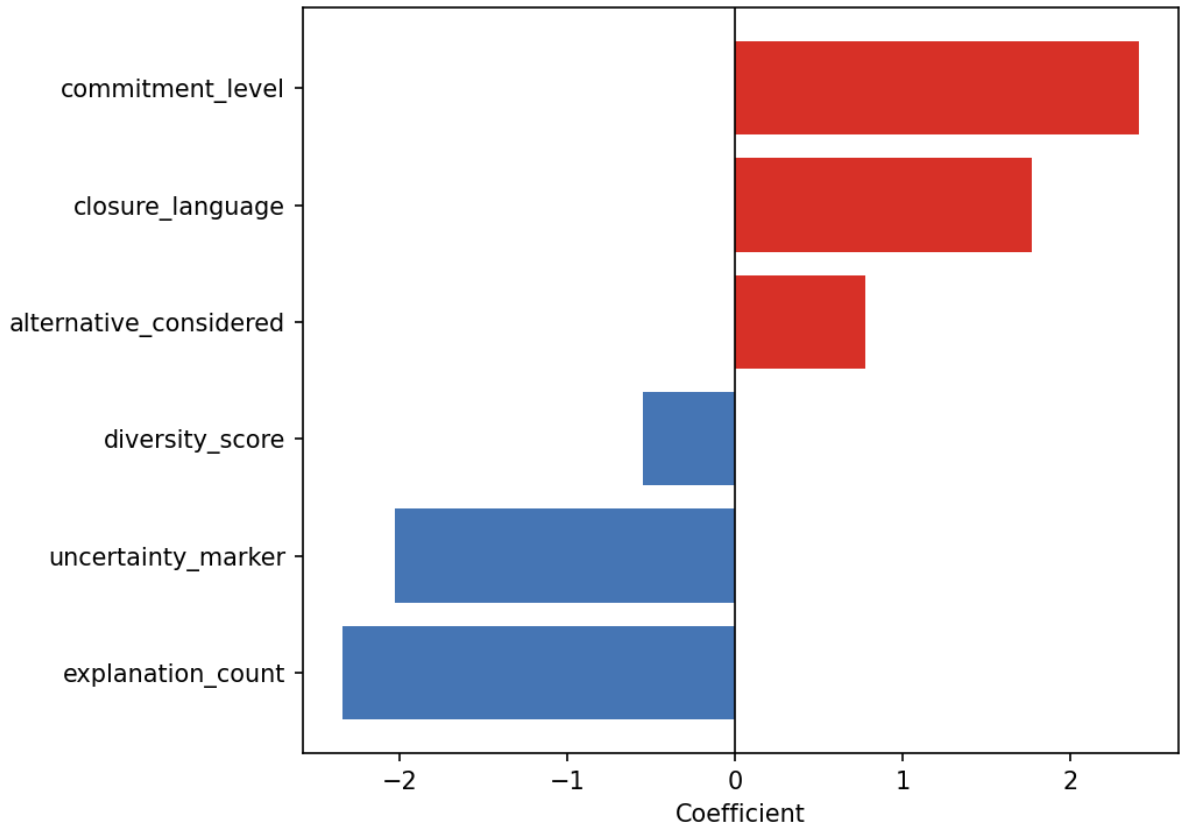


Figure 5: Feature importance coefficients from Logistic Regression. Red bars indicate features that increase premature convergence risk; blue bars indicate features that decrease risk.

Ablation Study

I conduct a leave-one-feature-out ablation study to assess individual feature contributions (Figure 6). Removing `commitment_level` produces the largest performance drop (ΔF1 = 0.060), identifying it as the most critical feature. Conversely, removing `diversity_score`, `closure_language`, and `alternative_considered` marginally improves performance, suggesting these features introduce noise under the current labeling scheme.

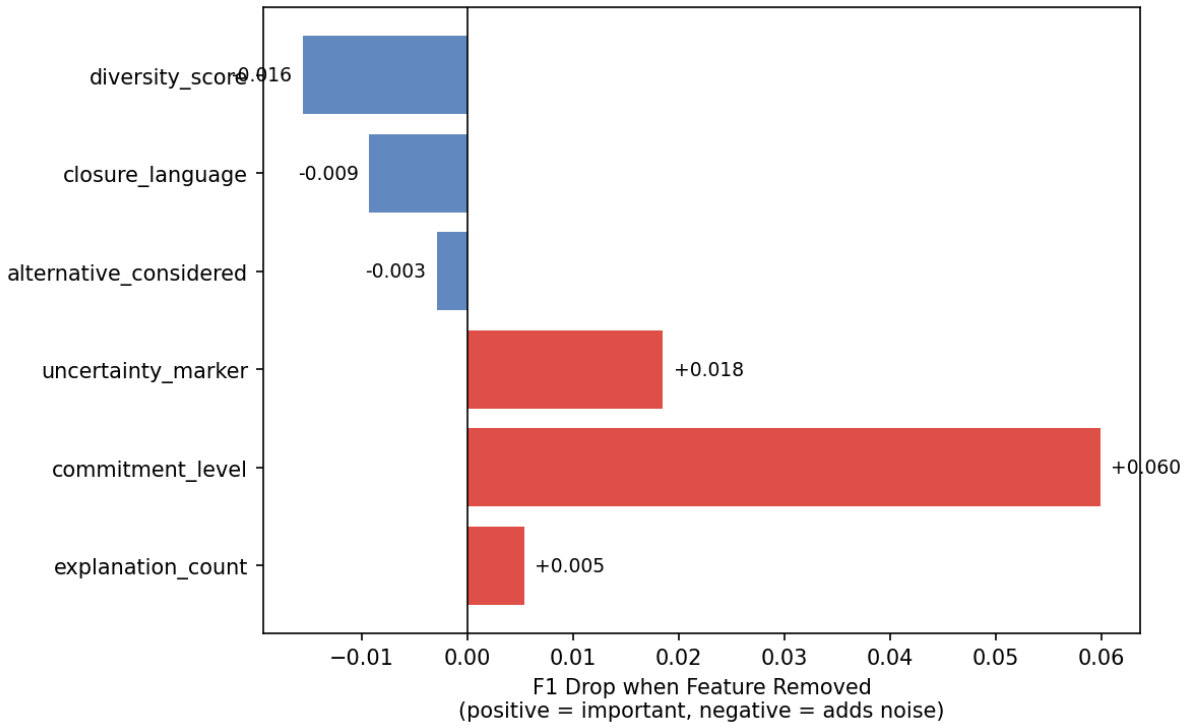


Figure 6: Ablation study: F1 change when each feature is removed. Red = important, Blue = adds noise.

SHAP Analysis

SHAP analysis (Figure 7) reveals that `explanation_count` and `diversity_score` contribute most strongly to individual predictions, while `commitment_level` operates as a necessary condition captured more effectively at the population level. This complementary pattern suggests that convergence is signaled through both epistemic commitment and explanatory breadth.

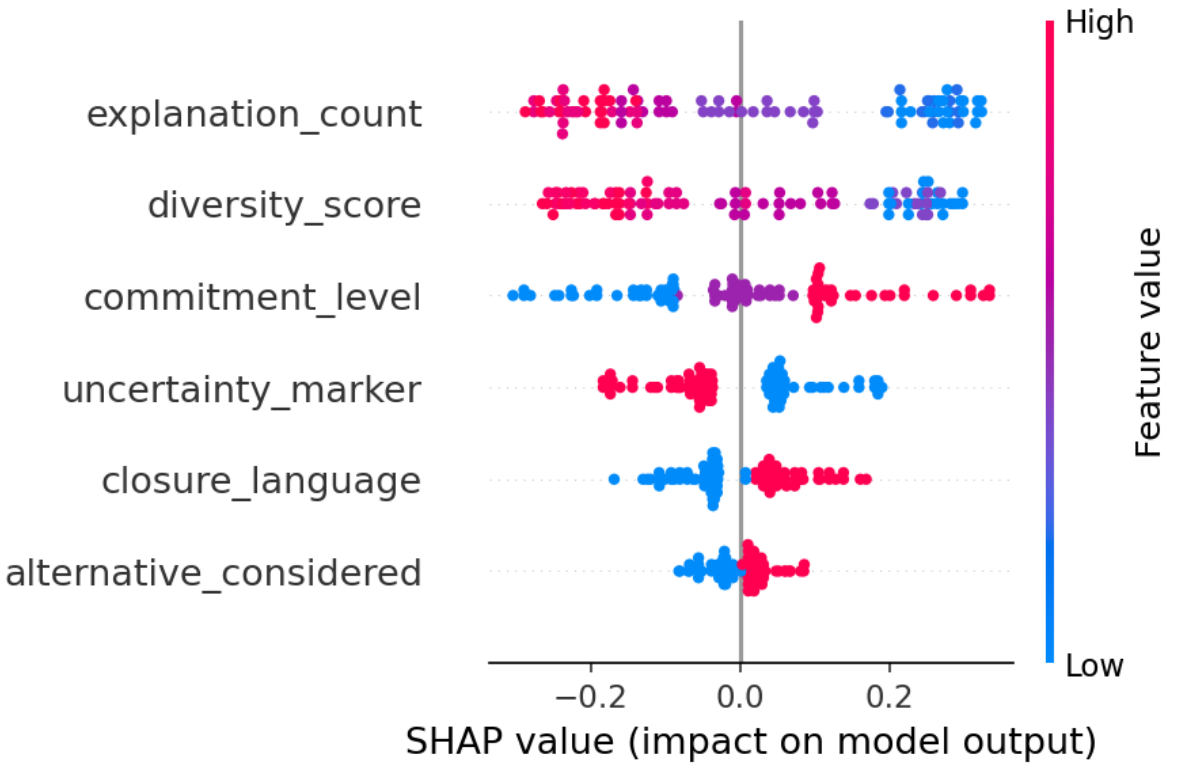


Figure 7: SHAP summary plot showing feature contributions to premature convergence prediction.

Discussion

Key Findings

Our results yield five findings relevant to accountability-aware AI design.

Finding 1: Commitment dominates at the population level. Ablation analysis identifies `commitment_level` as the strongest predictor (ΔF1 = 0.060), suggesting that epistemic certainty — not surface linguistic markers — is the primary signal of premature convergence.

Finding 2: Explanation count drives case-level variation. SHAP analysis reveals that `explanation_count` contributes most strongly to individual predictions, complementing the population-level finding. Together, these suggest a two-signal model: commitment as a necessary condition, explanation breadth as the discriminating factor.

Finding 3: Synthetic data approximates real-world patterns. The near-identical diversity scores between synthetic (1.584) and FDA Warning Letter corpus (1.601) data ($\Delta = 0.017$) support the ecological validity of our labeling design.

Finding 4: PAC is robustly detectable. AUC scores above 0.97 across three model architectures suggest that premature convergence leaves consistent, learnable traces in process-level features.

Finding 5: 24% of FDA Warning Letter cases show elevated convergence risk. Applying our risk score to the FDA Warning Letter corpus identifies 12 of 50 cases (24%) above the convergence threshold, with WL2018-05 exhibiting an outlier score of 1.0 — a single violation with zero explanatory diversity.

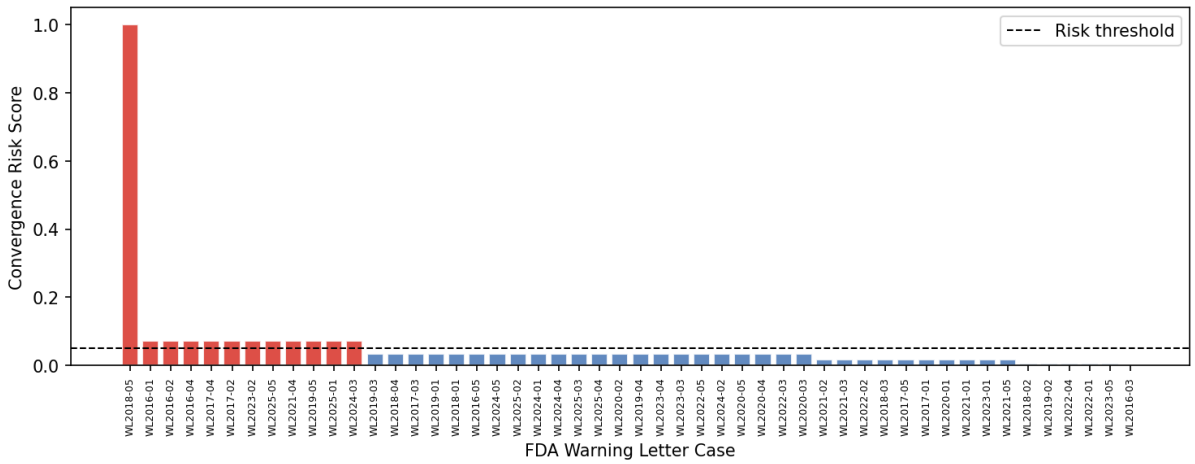


Figure 8: Premature convergence risk scores across 50 FDA Warning Letter cases. 12 cases (24%) exceed the risk threshold.

Limitations

Several limitations should be noted. First, our synthetic dataset relies on probabilistic labeling rather than human annotation, which may introduce systematic bias. Second, FDA Warning Letters represent final regulatory outputs rather than in-process reasoning logs, limiting our ability to observe convergence as it unfolds. Third, our feature set is hand-designed; future work should explore learned representations from transformer-based models. Additionally, the FDA

Warning Letter corpus exhibits limited responsibility subject diversity (Firm: 159/160 cases), reflecting the structural nature of regulatory enforcement documents rather than a limitation of the detection framework itself.

Implications for FAccT

Our framework connects to core FAccT concerns in two ways. First, premature convergence is itself a fairness issue: when accountability narrows too quickly onto individuals rather than systems, structural contributors go unexamined. Second, our detection approach provides a technical foundation for accountability auditing tools — systems that flag investigative closure before alternatives have been adequately explored.

Conclusion

I presented a computational framework for detecting premature accountability convergence in regulatory investigations. By operationalizing PAC as a process-quality classification task, I demonstrated that convergence leaves detectable traces in structural and linguistic features of investigative reasoning. Three classifiers achieve AUC above 0.97, with commitment level and explanation count emerging as the strongest predictive signals. Validation against an FDA Warning Letter corpus confirms the ecological validity of our synthetic design.

This work advances a computational approach to accountability that moves beyond outcome assessment toward process quality — asking not only whether the right answer was reached, but whether the reasoning remained sufficiently open before closing.

Future work will extend this framework to real-time reasoning logs, human annotation studies, and transformer-based feature extraction, with the goal of building accountability-aware investigation support systems for high-stakes regulatory environments.

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